

REVIEW

A review of established and new developments in local therapies for liver cancer

P. Seda¹, R. Abbey², L. Peng², I. Ezeaku³, S. T. Laroia⁴ & H. Aziz^{1*}

¹Department of Surgery, University of Iowa, University of Iowa Health Care, Iowa City; ²Carver College of Medicine, University of Iowa, Iowa City; ³Surgical Recovery and Perfusion Department, Gift of Hope Organ and Tissue Donor Network, Itasca; ⁴Department of Radiology, University of Iowa Health Care, Iowa City, USA

Available online xxx

Liver cancer, especially hepatocellular carcinoma (HCC), which accounts for ~75%-85% of primary liver cancers, remains a major global health challenge. Because of its high incidence, late diagnosis, and limited curative options, local therapies have become increasingly popular for the multidisciplinary management of HCC, especially in early and intermediate Barcelona Clinic Liver Cancer stages or in medically inoperable patients. This review examines a landscape of local treatment modalities for HCC encompassing radiofrequency ablation, microwave ablation, cryoablation, irreversible electroporation, histotripsy, transarterial chemoembolization, transarterial radioembolization, and stereotactic body radiotherapy. We also highlight their respective advantages and disadvantages, compare their survival outcomes, and identify current gaps in the literature, including the need for further comparisons between safety profiles and efficacy, and the growing landscape that is using different local therapies in a sequential or combinatorial fashion. Emphasis is placed on treatment decisions tailored to tumor burden, liver function, and patient-specific considerations. Expanding access to advanced local therapies in resource-limited settings remains a global priority.

Key words: liver cancer, radiotherapy, locoregional therapy, targeted treatment immunotherapy

INTRODUCTION

The incidence of primary liver cancer has been steadily rising worldwide, with hepatocellular carcinoma (HCC) representing ~75%-85% of primary liver malignancies.^{1,2} HCC often arises in the context of chronic liver diseases, including viral hepatitis, alcohol-related liver injury, or nonalcoholic fatty liver disease.³ In addition to primary tumors, the liver is a frequent site of metastatic spread from other primary cancers, such as colorectal, breast, gastric, and pancreatic cancers, due to its rich dual blood supply and unique microenvironment.⁴

Given the liver's anatomic accessibility and the advancements in imaging modalities such as ultrasound, computed tomography (CT), and magnetic resonance imaging (MRI), liver lesions are diagnosed and monitored with high precision.⁵ These same imaging techniques also facilitate minimally invasive local therapies, including radiofrequency ablation (RFA), microwave ablation (MWA), irreversible electroporation (IRE), and histotripsy, which are

often employed for patients who are not candidates for surgical resection.⁶ Because of its unique vascular supply, HCC can also be treated with transarterial infusion of anticancer treatments such as transarterial chemoembolization (TACE), transarterial radioembolization (TARE), and intra-arterial chemotherapy.^{7,8} These therapies allow for high concentrations of anticancer agents to be delivered directly to the tumor while minimizing systemic toxicity.⁹ The final locoregional therapy, stereotactic body radiotherapy (SBRT), delivers external-beam radiation to well-defined tumor sites. The advantages/disadvantages of each local therapy and the main comparative studies used throughout this review to compare the local therapies and their outcomes with each other are summarized in [Tables 1](#) and [2](#), respectively.

In recent years, there has been growing interest in combining locoregional treatments with systemic therapies, particularly tyrosine kinase inhibitors and immune checkpoint inhibitors. Emerging evidence suggests that **local therapies may enhance systemic immune responses** by promoting tumor antigen release and modulating the tumor microenvironment, thereby improving the efficacy of immunotherapy. Integrated treatment strategies that leverage both local and systemic modalities represent a promising frontier in the management of liver cancers.

In this review, we aim to comprehensively summarize current evidence on local therapies for HCC, focusing on

*Correspondence to: Dr Hassan Aziz, Department of Surgery, University of Iowa Health Care, 200 Hawkins Drive, C41-S GH, Iowa City, IA 52242, USA. Tel: (319) 467-5303
E-mail: Hassan-Aziz@uiowa.edu (H. Aziz).

2949-8198/© 2025 The Authors. Published by Elsevier Ltd on behalf of European Society for Medical Oncology. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

Table 1. Summary of the discussed local therapies

Modality	Advantages	Disadvantages	Future work needed
Resection	Most consistently effective treatment option and improved survival outcomes.	Potential post-operative complications and adverse side-effects; many patients are not candidates for this option; and expensive.	Minimally invasive alternatives and studies on the effects of surgery in combination with other local therapies.
RFA	Similar outcomes to resection for small tumors.	Heat-sink effect; not indicated for lesions that are in high-risk areas; and impaired visualization of the ablation field.	Uses when combined with other local therapies.
MWA	Similar outcomes to RFA and its mechanism combats the heat-sink effect.	Not indicated for lesions that are in high-risk areas and impaired visualization of the ablation field.	Evaluate whether RFA versus MWA is superior for certain cases and compare MWA with resection directly.
CRA	Less damage to surrounding tissues, which high temperatures (RFA/MWA) often cause. Clearer visualization of the ablation field. No heat-sink effect.	Risk for incomplete ablation.	More in-depth comparison to other modalities, such as resection, RFA, and MWA.
IRE	Technique combats the heat-sink effect; cells are damaged mainly via apoptosis versus necrosis; and may lead to lower tumor progression rates when compared with MWA/RFA.	Not indicated for lesions that are in high-risk areas and may have secondary heating effects that ultimately cause necrosis rather than apoptosis.	More in-depth comparison to other modalities such as resection and CRA.
Histotripsy	Cavitation technique combats the heat-sink effect. Treatment is predictable and not excessively ablative, with a good safety profile.	Further clinical trials are needed; besides, a few procedure-related major complications have been reported.	More in-depth comparison to other modalities such as resection, ablation, and IRE.
TACE	Sparers the systemic effects of chemotherapy and the surrounding liver tissue.	Risk for large-vessel occlusions and/or postembolization syndrome.	Evaluate whether cTACE or DEB-TACE is superior and the efficacy of combining TACE with ablation in comparison to resection.
TARE	Lower risk for adverse side-effects when compared with TACE.	Still a risk for large-vessel occlusions and/or postembolization syndrome.	Uses when combined with other local therapies, in comparison to resection.
SBRT	Sparers widespread effects of radiation, may be a suitable bridge to transplantation or definitive treatment for those ineligible.	Adverse effects from radiation.	More studies on its uses as a curative option in comparison to other first-line treatments, such as resection and ablation. Further studies on its uses as a combinatorial therapy, much like TACE.

CRA, cryoablation; cTACE, conventional TACE; DEB-TACE, drug-eluting bead TACE; IRE, irreversible electroporation; MWA, microwave ablation; RFA, radiofrequency ablation; SBRT, stereotactic body radiotherapy; TACE, transarterial chemoembolization; TARE, transarterial radioembolization.

both established and emerging techniques. We highlight recent clinical trials, compare efficacy and safety profiles, and discuss how these modalities may be integrated with systemic therapies such as tyrosine kinase inhibitors and immunotherapy. By clarifying the comparative benefits and limitations of local approaches, this review intends to guide future research directions and inform multidisciplinary treatment strategies for HCC.

METHODS

Literature review

A comprehensive review was conducted using Medline/PubMed, Google Scholar, Cochrane Library, and the Web of Science, which were accessed between 2000 and 2025. The main search focused on 'Local Therapy for Liver Cancer'. The following terms were used: liver cancer, radiotherapy, loco-regional therapy, ablative therapy, targeted treatment, and immunotherapy.

Study selection and data collection

Only English-language, full-text articles encompassing single case reports and original scientific research were included in this review. The reference lists of each article were examined to identify additional relevant studies for consideration. This review adhered to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses

guidelines. The initial articles were reviewed by PS and RA. These articles were confirmed via a review by HA.

Ethical compliance

All procedures carried out in this study involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards.

LOCAL THERAPIES

A wide range of locoregional therapeutic options have been developed for HCC, in part driven by the limited efficacy of systemic treatments.¹⁰ This limitation is partly due to the frequent coexistence of HCC with cirrhosis and its high molecular heterogeneity.¹¹ Surgical resection, transplantation, local ablation, and regional therapies have helped fill this gap, with improved survival outcomes confirmed by prospective trials in both primary and metastatic liver tumors.¹⁰

Surgical resection has been shown to improve survival outcomes and thus is the mainstay of curative therapy for primary liver tumors when amenable.¹⁰ However, a proportion of patients are not candidates for surgery due to the extent of the tumor or underlying poor liver function.¹² Surgical resection carries inherent risks, which are exacerbated by compromised liver function due to conditions

Table 2. Summary of studies mentioned that compared local therapies.

Comparison	References	Outcome
Resection versus RFA	17,18	RFA offered similar outcomes to resection for small tumors. RFA is more minimally invasive and cost-effective; resection is indicated more for high-risk nodules.
MWA versus RFA	22,24	MWA created larger ablation zones than RFA when treating tumors sized 1.5-4 cm; in the other study, MWA was more effective than single-needle RFA in reducing local tumor progression, but observed no significant differences between overall survival and recurrence-free survival at a 2-year follow-up.
Resection versus MWA versus RFA	28	Liver resection was more effective at achieving disease-free survival at 1 and 3 years when compared with MWA and RFA, with no significant difference in overall survival.
CRA versus RFA	28	CRA demonstrated no significant differences in efficacy when compared with RFA. The authors note that CRA was also not inferior to liver resection.
IRE versus MWA versus RFA	30	IRE demonstrated significantly lower tumor progression rates at 2 years when compared with RFA ($P = 0.005$), but no significant difference when compared with MWA.
cTACE versus DEB-TACE	39,40	There were no significant differences between the cTACE and DEB-TACE groups in terms of mortality rate, survival rate at 1-, 3-, and 5-year increments, nor in the frequency of adverse events. However, one significant difference was that the cTACE group did have a higher proportion of patients who achieved a complete radiological response when compared with the DEB-TACE group. However, neither of these differences had an impact on overall survival rates; in the other study, there were no significant differences between DEB-TACE and cTACE in terms of a complete or partial response rate.
RFA versus TACE+RFA	42,43	The sequential approach of TACE+RFA was statistically more effective than RFA alone in both overall survival rates and recurrence-free survival rates.
TACE+RFA versus resection	41	There were no significant differences between TACE+RFA and surgical resection in overall survival/recurrence-free survival rates at the 1-, 3-, and 5-year follow-ups. However, the incidence of complications was significantly lower in the TACE+RFA group versus the surgical resection group.
TARE versus TACE	48,49	No significant difference in overall survival or progression-free survival between the two modalities. TARE demonstrated a significantly longer time-to-progression compared with TACE; in the other study, while TACE and TARE appear to offer similar effectiveness in terms of overall survival, TARE may offer a heightened safety profile in patients with unresectable HCC.

CRA, cryoablation; cTACE, conventional TACE; DEB-TACE, drug-eluting bead TACE; HCC, hepatocellular carcinoma; IRE, irreversible electroporation; MWA, microwave ablation; RFA, radiofrequency ablation; TACE, transarterial chemoembolization; TARE, transarterial radioembolization.

such as cirrhosis or long-term systemic treatments.^{13,14} In such cases, the liver's reduced functional reserve limits its capacity to recover from tissue loss. Locoregional therapies provide a means of delivering tumor-directed cytotoxic effects while sparing surrounding nontumoral liver parenchyma.¹⁵

Ablative therapies

Tumor ablation is a locoregional therapy that can be applied with curative or palliative intent, either as a stand-alone treatment, in combination with other locoregional or surgical approaches, or as a bridge to liver transplantation. According to the Barcelona Clinic Liver Cancer (BCLC) staging system, ablation therapy is currently recommended for patients with early-stage disease (BCLC-0 or BCLC-A) who are not candidates for surgical resection or liver transplantation.¹⁶ These typically include patients with one to three nodules 3 cm in diameter and preserved liver function (Child–Pugh class A or B). Many modalities of ablation exist, with the two primary classes being thermal ablation [RFA, MWA, cryoablation (CRA), and photothermal therapy] and nonthermal ablation (IRE and photodynamic therapy).¹⁷

When compared with resection, ablation therapy may offer a similar survival outcome for early-stage disease (tumors size <3 cm) and is a more cost-effective, minimally invasive alternative.¹⁸

Radiofrequency ablation. This is the most widely used local ablative technique for the treatment of HCC. It employs alternating electrical current in the range of 375-480 kHz to

generate ionic agitation and frictional heat, producing tissue temperatures exceeding 60°C that result in coagulative necrosis under ultrasound, CT, or laparoscopic guidance.¹⁹ Despite its broad use and proven efficacy, RFA has several well-recognized limitations that may lead to incomplete tumor ablation. These include the 'heat-sink' effect caused by adjacent large blood vessels dissipating thermal energy; impaired visualization or targeting of very small (<1 cm) or subcapsular tumors; and reduced electrical conductivity in peritumoral tissue due to charring and tissue desiccation during treatment.

Microwave ablation. To combat the heat-sink effect, MWA was developed. MWA induces coagulative necrosis through electromagnetic waves—typically at frequencies of 915 MHz or 2.45 GHz—that heat tissue to temperatures exceeding 60°C.²⁰ As microwaves propagate through biological tissue with minimal dependence on electrical conductivity, MWA can achieve faster, deeper, and more homogenous heating than RFA, theoretically reducing the likelihood of incomplete ablation near large vessels. This theory is supported by a randomized, controlled, phase II trial from 2022 that compared MWA with RFA and observed that MWA created larger ablation zones than RFA ($P = 0.036$) when treating tumors sized 1.5-4 cm.²¹

In 2023, the Society of American Gastrointestinal and Endoscopic Surgeons (SAGES) and the Americas Hepato-Pancreato-Biliary Association (AHPBA) guidelines regarding the use of RFA versus MWA concluded that there is insufficient evidence to recommend one modality over the other in terms of oncological outcomes.²² A newer randomized controlled trial conducted after 2023

compared MWA versus single-needle RFA in HCC up to 4 cm and concluded that MWA was more effective than single-needle RFA in **reducing local tumor progression**. However, this study found **no significant differences between overall survival and recurrence-free survival at a 2-year follow-up**.²³

Cryoablation. This method induces tumor cell death through repeated rapid freeze–thaw cycles that generate intracellular ice crystals, osmotic stress, and microvascular disruption, ultimately leading to coagulative necrosis.²⁴ Benefits of CRA over RFA/MWA are less damage to vascular smooth muscle, blood vessels, bile ducts, and connective tissue, which are more susceptible to heat-induced damage. CRA also offers a clearer visualization of the ablation zone due to a visible ‘ice ball’ formation during the procedure.²⁴ RFA and MWA may be relatively contraindicated in patients with high-risk tumor locations—such as near the diaphragm, gastrointestinal tract, or major vessels—where poor visibility and the risk of thermal injury to adjacent organs are major concerns. In such settings, CRA may offer a safe and equally effective alternative. A 2024 study supported this hypothesis by demonstrating there was no significant difference in complete ablation rates or acquired complications between high- and low-risk lesion sites when using CRA.²⁵ However, a meta-analysis from 2024 that compared liver resection, RFA, MWA, and CRA for early-stage HCC across 25 publications observed that liver resection was more effective at achieving disease-free survival at 1 and 3 years when compared with MWA and RFA, with no significant difference in overall survival rates.²⁶ Of these 25 studies, only 2 (from 2008 and 2022) evaluated CRA and did so against RFA rather than liver resection. While CRA demonstrated no significant differences in efficacy when compared with other ablation modalities, the authors noted that CRA was not inferior to liver resection and emphasized the need for further randomized controlled trials comparing CRA with liver resection. Overall, CRA represents a valuable nonthermal ablation option, particularly for tumors in anatomically challenging or high-risk locations, although robust long-term and comparative data are still needed to define its precise role in HCC management.

Irreversible electroporation. This is a nonthermal ablative approach to treating HCC. Its fundamental mechanism involves the delivery of high-voltage electrical pulses that induce nanopore formation in cell membranes, triggering apoptosis rather than coagulative necrosis.²⁷ Theoretically, IRE has advantages over thermal ablation for lesions near vital structures or blood vessels as it avoids the ‘heat-sink’ effect. As it primarily disrupts cell membranes while preserving the extracellular matrix, vasculature, and bile ducts, IRE can maintain tissue architecture and minimize collateral injury. In practice, however, IRE produces a secondary heating effect during treatment, which may cause some collateral necrosis of the surrounding tissue. To evaluate IRE’s efficacy, a retrospective study from 2022 reviewed

322 patients with early-stage HCC who had undergone either IRE ($n = 15$), MWA ($n = 91$), or RFA ($n = 216$) treatment.²⁸ This investigation concluded that IRE demonstrated significantly lower tumor progression rates at 2 years when compared with RFA ($P = 0.005$), but no significant difference when compared with MWA. This conclusion is limited by the imbalances in sample size for each group. However, it suggests there is a need for further clinical trials on the efficacy of IRE in comparison to other established modalities, much like CRA. IRE thus represents a promising nonthermal ablation option for perivascular or peribiliary lesions, though its long-term oncologic efficacy remains under investigation.

Histotripsy. In 2023, the Food and Drug Administration (FDA) approved histotripsy, a noninvasive and nonthermal tissue ablation technique that utilizes ultrasound waves to target cancerous tissue.²⁹ Histotripsy poses a suitable alternative for patients with inoperable tumors or cirrhosis who are not candidates for resection, transplantation, or radiation. Histotripsy works by generating and collapsing microbubbles—known as acoustic cavitation clouds—within the targeted tissue, resulting in precise mechanical disruption and cellular liquefaction while sparing adjacent structures.³⁰ Variations in pulse duration, power, and frequency define different histotripsy regimens that can be optimized to control the extent and selectivity of ablation. The two hallmark human clinical trials studying histotripsy for HCC treatment are the THERESA study (NCT03741088), which established the foundation for histotripsy as a predictable ablation technique with a high safety profile, and the HOPE4LIVER (NCT04572633) study, which extrapolated on the THERESA findings.^{31,32} The THERESA study evaluated 11 tumors from eight patients with end-stage primary or secondary liver cancer.³¹

The THERESA study’s co-primary endpoints were achievement of technical success in 70% of patients, defined by concordance between targeted and treated tumor volume within 36 h after treatment (CT/MRI), and the incidence of 25% major procedure-related complications within 30 days. Histotripsy met and exceeded both endpoints, with 95% (42/44) achieving technical success and only 7% (3/44) experiencing major complications. These results directly supported FDA approval.³¹ The HOPE4LIVER study enrolled 44 participants across multiple countries and led to FDA approval of histotripsy after demonstrating and exceeding trial endpoint goals.³² The study’s first goal included meeting technical success in at least 70% of the patients. This was evaluated via CT or MRI 36 h after the procedure to see whether the tumor treatment volume encompassed the targeted volume. The second goal was to minimize the number of patients who experienced procedure-related complications within 30 days, with success deemed as complications in no more than 25% of the patients. Of the 44 patients, 95% (42/44) met the technical success endpoint within 36 h, and only 7% (3/44) experienced procedure-related major complications within 30 days. Some limitations of this study are the

small sample size, as well as using short-term post-procedure follow-up as an indication of successful treatment. Of note, in 2025's 1-year update, the HOPE4LIVER trial still demonstrates a favorable safety profile in terms of procedure-related adverse effects and reported an overall survival of 73.3% for patients with HCC and 48.6% for patients with metastatic disease.³³

INTRA-ARTERIAL THERAPIES

Patients with intermediate-stage HCC (BCLC-B) typically present with multifocal disease, preserved liver function, and no evidence of vascular invasion or extrahepatic spread.¹⁶ Selected BCLC patients with well-defined nodules may still meet the extended liver transplantation criteria, which offer the best curative chances. Patients who are not candidates for liver transplantation but have preserved portal flow to definable tumors may benefit from non-curative approaches such as TACE, TARE, or SBRT. These modalities aim to achieve locoregional tumor control, delay disease progression, and, in some cases, serve as a bridge to potentially curative interventions.

Transarterial chemoembolization

According to contemporary BCLC guidance, TACE is considered the first-line therapy for patients with intermediate-stage (BCLC-B) HCC who are candidates for liver-directed treatment. In early-stage disease (BCLC-0/A), TACE may be used when curative options (resection or ablation) are not feasible, have failed, or as a bridge to transplantation.¹⁶ TACE is a locoregional therapy that delivers a high concentration of chemotherapy into the tumor-feeding arteries followed by embolization, thereby inducing ischemia of the tumor while limiting systemic exposure.³⁴ This selectivity leverages the predominantly arterial blood supply of HCC compared with the largely portal venous perfusion of normal liver parenchyma.³⁵ There are two main TACE modalities: conventional TACE (cTACE) and drug-eluting bead TACE (DEB-TACE). cTACE involves the injection of a cytotoxic agent (such as doxorubicin or cisplatin), followed by the infusion of the oil-based and radio-opaque chemotherapeutic carrier lipiodol, and concludes with the injection of an embolic material.³⁶ The commonly used embolic materials are degradable starch microspheres, collagen, and gelatin sponge. In contrast, DEB-TACE uses nonresorbable embolic microspheres that release chemotherapeutic drugs in a prolonged manner.

Comparative studies of cTACE versus DEB-TACE have yielded mixed results. A 2024 analysis retrospectively evaluated 268 patients who underwent cTACE ($n = 198$) or DEB-TACE ($n = 70$) treatment for HCC between January 2009 and December 2021.³⁷ There were no significant differences between the cTACE and DEB-TACE groups in terms of mortality rate; survival rate at 1-, 3-, and 5-year increments; or frequency of adverse events. However, the cTACE group did have a significantly higher proportion of patients who achieved a complete radiological response

when compared with the DEB-TACE group, which instead demonstrated a higher proportion of partial response rates when compared with the cTACE group. A larger meta-analysis from 2021 evaluated data from 34 studies, comparing 4841 patients who underwent either cTACE treatment ($n = 2558$) or DEB-TACE treatment ($n = 2283$).³⁸ In this analysis, there were no significant differences between DEB-TACE and cTACE in terms of complete or partial response rate. This study suggested that DEB-TACE was superior in terms of safety outcomes and disease control, although the differences were not statistically significant. Overall, evidence remains conflicting with low-to-moderate certainty; therefore further high-quality trials are needed.

Uniquely, TACE can also be combined with other locoregional therapies such as RFA (TACE–RFA). One theoretical benefit of this combination is the ability to minimize the heat-sink effect of RFA by occluding hepatic blood flow with TACE.³⁹ Previous studies have demonstrated that a sequential TACE–RFA approach was statistically superior to RFA alone for both overall survival rates and recurrence-free survival rates.^{40,41} A 2025 trial of 210 patients with small late-recurrence HCC investigated treatment outcomes between patients receiving TACE–RFA and surgical resection.³⁹ Ultimately, no significant differences were found between the two groups in overall survival or recurrence-free survival rates at 1-, 3-, and 5-year follow-ups. This study did find that the incidence of complications was significantly lower in the TACE–RFA group than in the surgical resection group.

Transarterial radioembolization

TARE (also known as selective internal radiation therapy) is not incorporated as a formal recommendation in the 2022 BCLC treatment algorithm; nevertheless, it is widely used across selected BCLC stages in clinical practice, including patients with unresectable disease and in scenarios less amenable to TACE.⁴² In March 2021, the United States FDA approved TheraSphere yttrium-90 (⁹⁰Y) glass microspheres for the treatment of unresectable HCC.^{43,44} TARE entails selective intra-arterial delivery of ⁹⁰Y-loaded microspheres to tumor-feeding arteries.⁴⁴ In contrast to TACE, embolization is not the primary therapeutic goal; rather, ⁹⁰Y delivers high-energy radiation with limited tissue penetration, achieving tumoricidal doses while preserving surrounding parenchyma.⁴⁵ Radioactive decay of ⁹⁰Y is independent of tissue oxygen, though oxygen can enhance radiobiologic cell-kill.⁴⁶

A meta-analysis from 2022 evaluated 17 studies of 2465 patients that underwent either TACE ($n = 1657$) or TARE ($n = 808$) for HCC treatment.⁴⁷ Of these patients, BCLC-B was the most common stage (42.1%), followed by BCLC-C (29.0%) and BCLC-A (30.3%). This meta-analysis found no significant difference in overall survival between the two modalities. Three of the analyses demonstrated that TARE had a significantly longer time-to-progression compared with TACE. To assess these modalities in larger tumors, a 2024 study evaluated TACE and TARE in patients with HCC

lesions >8 cm.⁴⁶ No significant differences in overall survival or progression-free survival were noted between the two modalities. Of note, appropriate propensity-score matching analysis revealed that all patients within the TACE group ($n = 40$) experienced postembolization syndrome, which accounted for 171 adverse events. In comparison, 30 patients within the TARE group ($n = 40$) experienced postembolization syndrome, which accounted for 72 adverse events. When adverse events were stratified by severity, 29 (72.5%) patients in the TACE group experienced a major adverse event, compared with 2 (5%) patients in the TARE group. While TARE and TACE demonstrate similar effects on overall survival in comparative studies, TARE often yields longer time-to-progression and fewer adverse events, particularly less postembolization syndrome. High-quality randomized trials are still needed to define optimal selection across BCLC stages.

Stereotactic body radiotherapy

Recent technical advances in radiation therapy have enabled the delivery of high radiation doses to focal tumors. This has led to the widespread use of SBRT for various masses, including in the lung, liver, and bone. SBRT delivers highly conformal external-beam radiation in a short time to well-defined tumor sites.⁴⁸ In early-stage HCC patients who are ineligible for surgical resection or thermal ablation, SBRT achieves durable local control and competitive survival outcomes. Across prospective cohorts and multi-institutional series, 2-3-year local control commonly approaches 90%-100% and overall survival ~60%-70%, albeit with heterogeneity in dose, fractionation, and selection criteria.⁴⁹⁻⁵²

Currently, SBRT is considered a curative alternative for patients who are ineligible for transplant surgery, ablation, TACE, or other systemic therapy. SBRT may also serve as a bridge to transplantation.⁴⁸ In addition, the role of SBRT has expanded beyond early-stage HCC to include cases with portal vein or inferior vena cava thrombi.⁵³ Further research is needed to solidify its role in clinical settings.⁵⁴

Combination treatment approaches

There are multiple theoretical advantages to combining locally ablative therapies with transarterial approaches such as TACE. By leveraging complementary mechanisms of action, combination strategies may potentiate antitumor efficacy and improve local control. When used in combination, dose and energy profiles for each locally ablative modality may be reduced in selected cases while maintaining cytotoxic effect, potentially widening the safety margin for adjacent normal tissues. Combination therapy can also mitigate modality-specific limitations—for example, reducing perfusion-related heat loss during thermal ablation.

One treatment combination, which has been extensively investigated in recent years, is the combination of TACE and thermal tumor ablation. This combination may decrease TACE-induced neo-angiogenesis and therefore reduce the

risk of tumor recurrence and metastatic growth.^{55,56} The combination of RFA and TACE also increases the ablation (coagulation) zone and has been associated with lower local tumor progression compared with either modality alone.⁵⁷ In early-stage, nonresectable HCC, multiple meta-analyses report improved overall survival and recurrence-free survival with TACE-plus-ablation versus monotherapy, with the largest benefits typically observed in larger tumors (e.g. >3-5cm).^{17,58} For intermediate-stage disease, ablation may be an appropriate adjunct to TACE when tumor location and accessibility are favorable.⁵⁸

At present, the strongest body of evidence supports RFA in combination with TACE. The biologic rationale is that arterial flow reduction after TACE limits conductive heat loss, improves ablation margins, may enhance control of satellite lesions, and has been associated with lower local tumor progression compared with either modality alone.⁵⁸

CONCLUSION

Local therapies play a critical role in managing HCC, particularly for patients ineligible for surgery. Ablation offers a minimally invasive alternative for small tumors, with RFA being the most common but limited by heat-sink effects. MWA provides larger ablation zones, while CRA preserves nearby tissues and is suitable for high-risk sites. Nonthermal methods such as IRE and histotripsy show promise near vital structures and in inoperable cases, though further trials are needed.

For intermediate-stage HCC, intra-arterial therapies such as TACE and TARE remain central. TACE is the standard first-line therapy, while TARE offers similar survival with fewer severe complications. SBRT is an emerging option, particularly as a bridge or alternative therapy.

A multidisciplinary approach and combination strategies, including integration with immunotherapy, may enhance outcomes. Future trials comparing novel ablation techniques, optimizing combinations, and refining patient selection will shape the next era of liver cancer treatment.

FUNDING

None declared.

DISCLOSURE

The authors have declared no conflicts of interest.

REFERENCES

1. Rumgay H, Ferlay J, de Martel C, et al. Global, regional and national burden of primary liver cancer by subtype. *Eur J Cancer*. 2022;161:108-118.
2. Orcutt ST, Anaya DA. Liver resection and surgical strategies for management of primary liver cancer. *Cancer Control*. 2018;25(1):1073274817744621.
3. Younossi ZM, Wong G, Anstee QM, Henry L. The global burden of liver disease. *Clin Gastroenterol Hepatol*. 2023;21(8):1978-1991.
4. Li X, Ramadori P, Pfister D, et al. The immunological and metabolic landscape in primary and metastatic liver cancer. *Nat Rev Cancer*. 2021;21(9):541-557.

5. Vernuccio F, Cannella R, Bartolotta TV, et al. Advances in liver US, CT, and MRI: moving toward the future. *Eur Radiol Exp*. 2021;5(1):52.
6. Izzo F, Granata V, Grassi R, et al. Radiofrequency ablation and microwave ablation in liver tumors: an update. *Oncologist*. 2019;24(10):e990-e1005.
7. Şener N, Yakupoğlu A. Yttrium-90 transarterial radioembolization and capecitabine in hepatocellular carcinoma with portal vein involvement. *Medicine*. 2023;102(35):e34674.
8. Wright GP, Perkins S, Jones H, et al. Surgical resection does not improve survival in multifocal intrahepatic cholangiocarcinoma: a comparison of surgical resection with intra-arterial therapies. *Ann Surg Oncol*. 2018;25(1):83-90.
9. Schuster R, Lindner M, Wacker F, et al. Transarterial chemoembolization of liver metastases from uveal melanoma after failure of systemic therapy: toxicity and outcome. *Melanoma Res*. 2010;20(3):191-196.
10. Petrowsky H, Fritsch R, Guckenberger M, et al. Modern therapeutic approaches for the treatment of malignant liver tumours. *Nat Rev Gastroenterol Hepatol*. 2020;17(12):755-772.
11. Lu L-C, Hsu CH, Hsu C, Cheng AL. Tumor heterogeneity in hepatocellular carcinoma: facing the challenges. *Liver Cancer*. 2016;5(2):128-138.
12. Vitale A, Huo TL, Cucchetti A, et al. Survival benefit of liver transplantation versus resection for hepatocellular carcinoma: impact of MELD score. *Ann Surg Oncol*. 2015;22(6):1901-1907.
13. Nathan H, Segev DL, Mayo SC, et al. National trends in surgical procedures for hepatocellular carcinoma: 1998-2008. *Cancer*. 2012;118(7):1838-1844.
14. Tanabe G, Sakamoto M, Akazawa K, et al. Intraoperative risk factors associated with hepatic resection. *Br J Surg*. 1995;82(9):1262-1265.
15. Li D, Kang J, Golas BJ, Yeung VW, Madoff DC. Minimally invasive local therapies for liver cancer. *Cancer Biol Med*. 2014;11(4):217-236.
16. Reig M, Forner A, Rimola J, et al. BCLC strategy for prognosis prediction and treatment recommendation: the 2022 update. *J Hepatol*. 2022;76(3):681-693.
17. Wang K, Wang C, Jiang H, et al. Combination of ablation and immunotherapy for hepatocellular carcinoma: where we are and where to go. *Front Immunol*. 2021;12:792781.
18. Cucchetti A, Piscaglia F, Cescon M, et al. Cost-effectiveness of hepatic resection versus percutaneous radiofrequency ablation for early hepatocellular carcinoma. *J Hepatol*. 2013;59(2):300-307.
19. Hinshaw JL, Lubner MG, Ziemlewicz TJ, et al. Percutaneous tumor ablation tools: microwave, radiofrequency, or cryoablation—what should you use and why? *Radiographics*. 2014;34(5):1344-1362.
20. Tang L, Lui H, Cheng J, Li J, Li Q. Cryoablation versus microwave ablation in the treatment of hepatocellular carcinoma: a systematic review and meta-analysis. *Asian J Surg*. 2025;48(3):1626-1633.
21. Radosevic A, Quesada R, Serlavos C, et al. Microwave versus radiofrequency ablation for the treatment of liver malignancies: a randomized controlled phase 2 trial. *Sci Rep*. 2022;12(1):316.
22. Ceppa EP, Collings AT, Abdalla M, et al. SAGES/AHPBA guidelines for the use of microwave and radiofrequency liver ablation for the surgical treatment of hepatocellular carcinoma or colorectal liver metastases less than 5 cm. *Surg Endos*. 2023;37(12):8991-9000.
23. Sugimoto K, Imajo K, Kuroda H, et al. Microwave ablation vs. single-needle radiofrequency ablation for the treatment of HCC up to 4 cm: a randomized-controlled trial. *JHEP Rep*. 2024;7(1):101269.
24. Pusceddu C, Mascia L, Ninniri C, et al. The increasing role of CT-guided cryoablation for the treatment of liver cancer: a single-center report. *Cancers*. 2022;14(12):3018.
25. Zhang W, Wang Y, Zhao X, et al. Efficacy and safety of CT-guided percutaneous cryoablation for hepatocellular carcinoma at high-risk sites. *Acad Radiol*. 2024;31(11):4434-4444.
26. Min Y, Tong K, Lin H, et al. Ablative treatments and surgery for early-stage hepatocellular carcinoma: a network meta-analysis. *J Surg Res*. 2024;303:587-599.
27. Vogel JA, van Veldhuisen E, Agnass P, et al. Time-dependent impact of irreversible electroporation on pancreas, liver, blood vessels and nerves: a systematic review of experimental studies. *PLoS One*. 2016;11(11):e0166987.
28. Wada T, Sugimoto K, Sakamaki K, et al. Comparisons of radiofrequency ablation, microwave ablation, and irreversible electroporation by using propensity score analysis for early stage hepatocellular carcinoma. *Cancers*. 2023;15(3):732.
29. Verma Y, Perera Molligoda Arachchige AS. Advances in tumor management: harnessing the potential of histotripsy. *Radiol Imaging Cancer*. 2024;6(3):e230159.
30. Queen H, Ferris SF, Cho CS, Ganguly A. The emerging role of histotripsy in liver cancer treatment: a scoping review. *Cancers*. 2025;17(6):915.
31. Vidal-Jove J, Serres X, Vlaisavljevich E, et al. First-in-man histotripsy of hepatic tumors: the THERESA trial, a feasibility study. *Int J Hyperthermia*. 2022;39(1):1115-1123.
32. Mendiratta-Lala M, Wiggemann P, Pech M, et al. The #HOPE4LIVER single-arm pivotal trial for histotripsy of primary and metastatic liver tumors. *Radiology*. 2024;312(3):e233051.
33. Ziemlewicz TJ, Critchfield JJ, Mendiratta-Lala M, et al. The #HOPE4-LIVER single-arm pivotal trial for histotripsy of primary and metastatic liver tumors: one-year update of clinical outcomes. *Ann Surg*. 2025;282(6):908-916.
34. Chang Y, Jeong SW, Young Jang J, Jae Kim Y. Recent updates of transarterial chemoembolization in hepatocellular carcinoma. *Int J Mol Sci*. 2020;21(21):8165.
35. Tsurusaki M, Murakami T. Surgical and locoregional therapy of HCC: TACE. *Liver Cancer*. 2015;4(3):165-175.
36. Kotsifa E, Vergadis C, Vailas M, et al. Transarterial chemoembolization for hepatocellular carcinoma: why, when, how? *J Pers Med*. 2022;12(3):436.
37. Fontana PC, Coral GP, Horbe AF, et al. Retrospective analysis of the efficacy and survival associated with cTACE and DEB-TACE in the palliative treatment of hepatocellular carcinoma: experience of a tertiary care hospital in southern Brazil. *Radiol Bras*. 2024;57:e20230105.
38. Bzeizi KI, Arabi M, Jamshidi N, et al. Conventional transarterial chemoembolization versus drug-eluting beads in patients with hepatocellular carcinoma: a systematic review and meta-analysis. *Cancers*. 2021;13(24):6172.
39. Zhang Y-J, Chen J, Zhou Z, et al. Transarterial chemoembolization with radiofrequency ablation versus surgical resection for small late-recurrence hepatocellular carcinoma. *Radiology*. 2025;314(2):e241096.
40. Peng Z-W, Zhang YJ, Liang HH, Lin XJ, Guo RP, Chen MS. Recurrent hepatocellular carcinoma treated with sequential transcatheter arterial chemoembolization and RF ablation versus RF ablation alone: a prospective randomized trial. *Radiology*. 2012;262(2):689-700.
41. Peng Z-W, Zhang YJ, Chen MS, et al. Radiofrequency ablation with or without transcatheter arterial chemoembolization in the treatment of hepatocellular carcinoma: a prospective randomized trial. *J Clin Oncol*. 2013;31(4):426-432.
42. Helmberger T, Golfieri R, Pech M, et al. Clinical application of transarterial radioembolization in hepatic malignancies in Europe: first results from the prospective multicentre observational study CIRSE registry for SIR-spheres therapy (CIRT). *Cardiovasc Intervent Radiol*. 2021;44(1):21-35.
43. Guir B, Garin E, Allimant C, Edeline J, Salem R. TARE in hepatocellular carcinoma: from the right to the left of BCLC. *Cardiovasc Intervent Radiol*. 2022;45(11):1599-1607.
44. Hamad A, Aziz H, Kamel IR, Diaz DA, Pawlik TM. Yttrium-90 radioembolization: current indications and outcomes. *J Gastrointest Surg*. 2023;27(3):604-614.
45. Fidelman N, Kerlan RK Jr. Transarterial chemoembolization and 90Y radioembolization for hepatocellular carcinoma: review of current applications beyond intermediate-stage disease. *Am J Roentgenol*. 2015;205(4):742-752.

46. Phan NH, Chun HJ, Oh JS, et al. TACE vs. TARE for HCC ≥ 8 cm: a propensity score analysis. *Abdom Radiol.* 2024;50(3):1198-1208.
47. Brown AM, Kassab I, Massani M, et al. TACE versus TARE for patients with hepatocellular carcinoma: overall and individual patient level meta analysis. *Cancer Med.* 2023;12(3):2590-2599.
48. Andolino DL, Johnson CS, Maluccio M, et al. Stereotactic body radiotherapy for primary hepatocellular carcinoma. *Int J Radiat Oncol Biol Phys.* 2011;81(4):e447-e453.
49. Seo YS, Kim MS, Yoo SY, et al. Preliminary result of stereotactic body radiotherapy as a local salvage treatment for inoperable hepatocellular carcinoma. *J Surg Oncol.* 2010;102(3):209-214.
50. Kwon JH, Bae SH, Kim JY, et al. Long-term effect of stereotactic body radiation therapy for primary hepatocellular carcinoma ineligible for local ablation therapy or surgical resection. Stereotactic radiotherapy for liver cancer. *BMC Cancer.* 2010;10:475.
51. Sanuki N, Takeda A, Oku Y, et al. Stereotactic body radiotherapy for small hepatocellular carcinoma: a retrospective outcome analysis in 185 patients. *Acta Oncol.* 2014;53(3):399-404.
52. Kubo K, Kimura T, Aikata H, et al. Long-term outcome of stereotactic body radiotherapy for patients with small hepatocellular carcinoma. *Hepatol Res.* 2018;48(9):701-707.
53. Kimura T, Fujiwara T, Kameoka T, Adachi Y, Kariya S. The current role of stereotactic body radiation therapy (SBRT) in hepatocellular carcinoma (HCC). *Cancers.* 2022;14(18):4383.
54. Hernandez L, Parent L, Molinier V, et al. Stereotactic body radiation therapy in primary liver tumor: local control, outcomes and toxicities. *Clin Transl Radiat Oncol.* 2025;50:100892.
55. Li W, Ni C-F. Current status of the combination therapy of transarterial chemoembolization and local ablation for hepatocellular carcinoma. *Abdom Radiol.* 2019;44(6):2268-2275.
56. Morimoto M, Numata K, Kondou M, et al. Midterm outcomes in patients with intermediate-sized hepatocellular carcinoma: a randomized controlled trial for determining the efficacy of radiofrequency ablation combined with transcatheter arterial chemoembolization. *Cancer.* 2010;116(23):5452-5460.
57. Kitamoto M, Imagawa M, Yamada H, et al. Radiofrequency ablation in the treatment of small hepatocellular carcinomas: comparison of the radiofrequency effect with and without chemoembolization. *Am J Roentgenol.* 2003;181(4):997-1003.
58. Li L, Tian J, Liu P, et al. Transarterial chemoembolization combination therapy vs monotherapy in unresectable hepatocellular carcinoma: a meta-analysis. *Tumori.* 2016;102(3):301-310.